ABC Inc.

Data Analysis Report

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Executive Summary

Overview

This report analyzes 16,842 logistics orders from ABC Inc. to evaluate delivery performance, isolate process bottlenecks, and optimize operational decision-making. Using SQL for data cleaning, interval calculation, feature engineering, and median imputation, key metrics were analyzed alongside Power BI correlation models to assess delivery efficiency.

Key Findings

- **Fulfillment Performance:** Out of 16,842 orders, 14,276 (84.76%) were completed, while 2,566 (15.24%) were cancelled. Among fulfilled orders, 7,906 were on time, 5,396 were late, and 902 were early.
- **Pickup Bottlenecks:** For late deliveries, the median total service time reached 106.58 minutes. Over 65% of this delay was due to the pickup phase, with a median pickup duration of 71.34 minutes.
- **Distance vs. Service Time Disconnect:** Total service time showed a weak correlation with total distance (R = 0.22). Furthermore, pickup duration and rider-to-store distance showed virtually no correlation (R = 0.05). Delays are driven primarily by operational inefficiencies rather than geographical distance.
- **Costly Order Cancellations:** 2,522 orders were cancelled at Stage 0 (placed), 40 at Stage 1 (prepared), and 4 at Stage 2 (picked up). Progressively higher stages represent severe sunk costs in inventory, labor, and fuel.
- **Data System Integrity:** Missing timestamps and untagged order statuses indicate rider reporting omissions or system logging bugs.

Strategic Recommendations

1. **Optimize Order Dispatched & Pickup Systems:** Implement dynamic rider dispatching and automated "Ready for Pickup" store notifications to eliminate idle waiting times. Enforce GPS-enabled rider check-ins.
2. **Mitigate Cancellation Costs:** Provide real-time order tracking to manage customer expectations. Conduct root-cause investigations for Stage 2 cancellations and consider implementing late-stage cancellation fees to recover sunk costs.
3. **Enhance System Data Governance:** Make timestamp logging mandatory in the app prior to order finalization, and audit system bugs responsible for untagged statuses.
4. **Implement Real-Time Operational Dashboards:** Shift from static analysis to dynamic tracking via Power BI to monitor fulfillment rates, fulfillment funnels, and pickup delays in real time.

Scope and Objectives

This report aims to analyze logistics data to improve delivery operations and corporate decision-making. By the end of the report, full details of the recommendations will be provided.

Methodologies

Data Definition

Column Header	Data Type	Description
order_id	Varchar	Unique identifier for each customer order.
order_status	Integer	Code representing the current state of the order (20, 30).
distance_to_customer	Float	Distance (km) from the store to the customer's delivery address.
distance_rider_from_store	Float	Current distance (km) between the rider and the store
delivery_service_level	Integer	Category or tier of delivery service (1, 2, 3).
rider_arrived_at_pickup_time	Datetime	Time indicating when the rider arrived at the store.
order_ready_for_pickup_time	Datetime	The time when the order was prepared and available for pickup.
order_pickup_done_time	Datetime	The actual time the rider collected the order from the store.
order_delivered_time	Datetime	The time when the order was successfully delivered to the customer.

Before diving into analysis, the data is cleaned of null values and inconsistent formatting through SQL. The date and time were converted to date intervals to measure the minutes between activities. Null values for each float data type were converted into the median values of their respective columns, since the data distribution is skewed to the right.

Feature engineering is also performed, adding various columns and replacing values or data descriptions that may introduce vagueness:

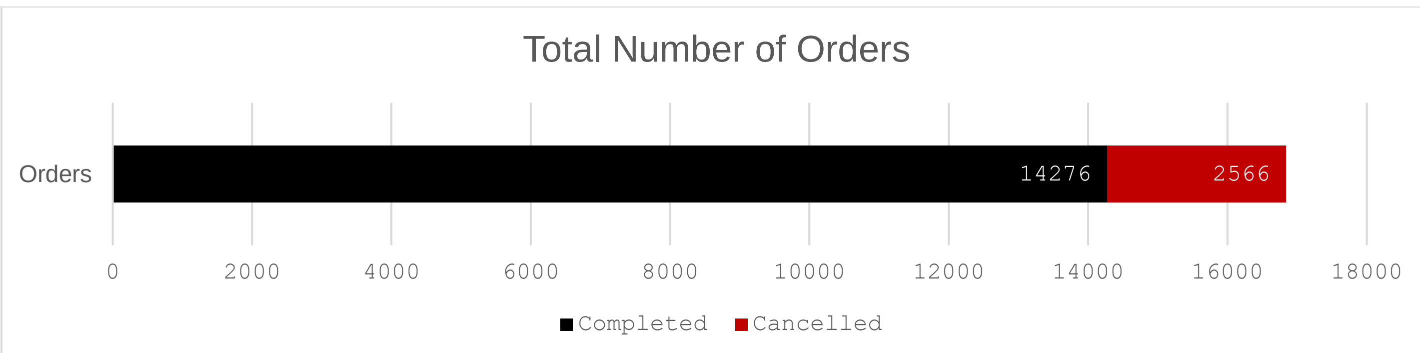
Column Header	Data Type	Description
order_id	Varchar	Unique identifier for each customer order.
delivery_performance	Varchar	Code (20, 30) converted into (Completed, Cancelled)
rider_store_km	Float	Current distance (km) between the rider and the store
store_customer_km	Float	Distance (km) from the store to the customer's delivery address.
total_dist_km	Integer	rider_store_km + store_customer_km Total distance traveled by the rider
mins_taken_to_pickup	Datetime	order_pickup_done_time – order_ready_for_pickup_time Indicates how long the rider took to get from their initial position to the store
mins_taken_to_deliver_from_pickup	Datetime	order_delivered_time – order_pickup_done_time

Column Header	Data Type	Description
		Indicates how long the rider took to deliver the product to the customer after picking the product up from the store
mins_total_service	Datetime	order_delivered_time – order_ready_for_pickup_time Indicates how long it took to deliver the product to the customer in minutes after the order is prepared
order_date	Datetime	The date and time when the order was prepared

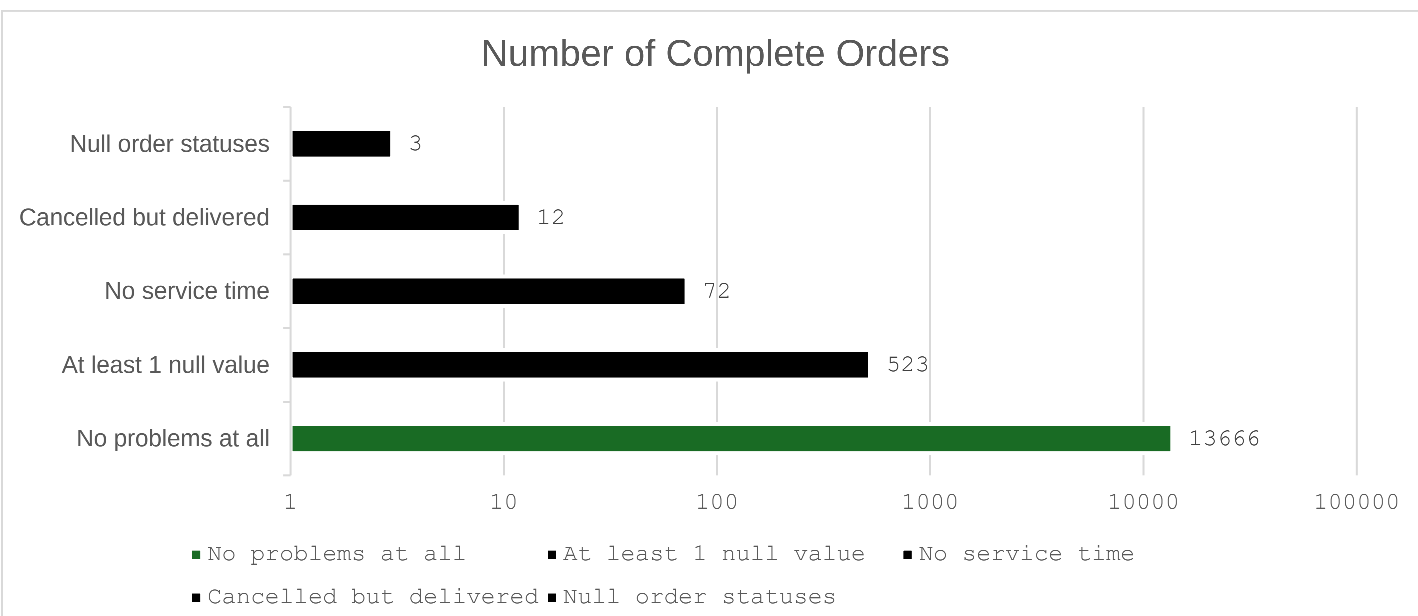
Results and Discussions

Description of the Orders

The logistics dataset contained 16,842 orders, of which 14,276 (84.76%) were fulfilled, and 2,566 (15.24%) were cancelled, as observed in the stacked bar chart below.

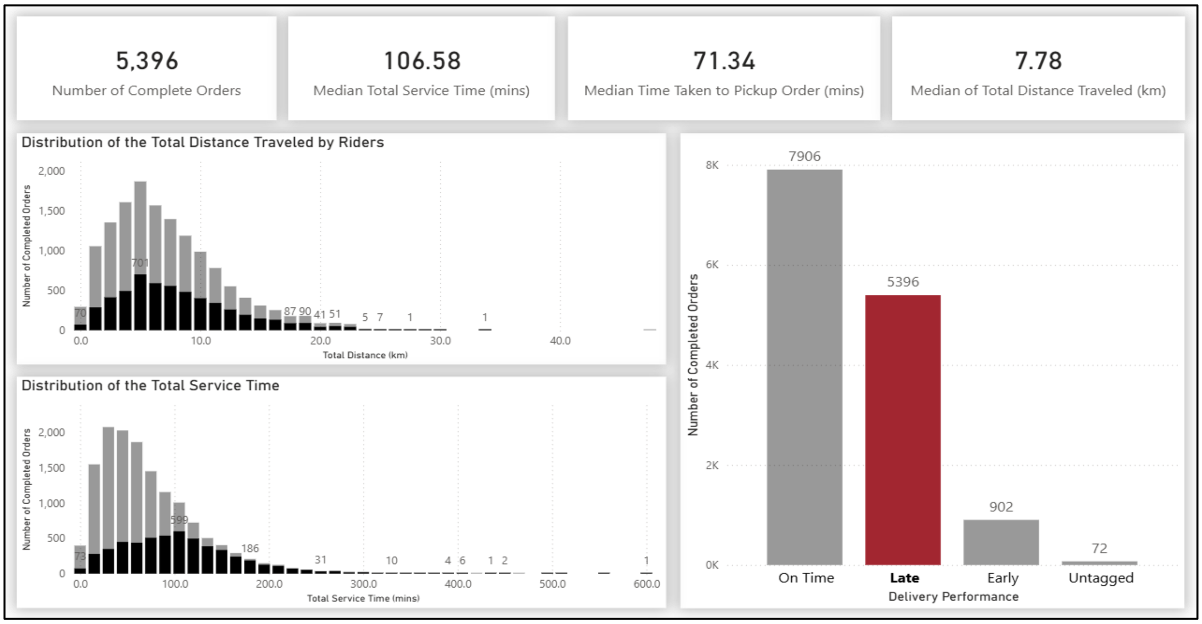


Within the completed orders, some values are missing and read as null. It is possible that the rider has forgotten to record the time they arrived at the store, picked up the order, or even delivered the product. It is also possible that they delivered the product but forgot to mark the order as completed, resulting in null order statuses. It may also mean that the rider did everything right, but the system is defective.

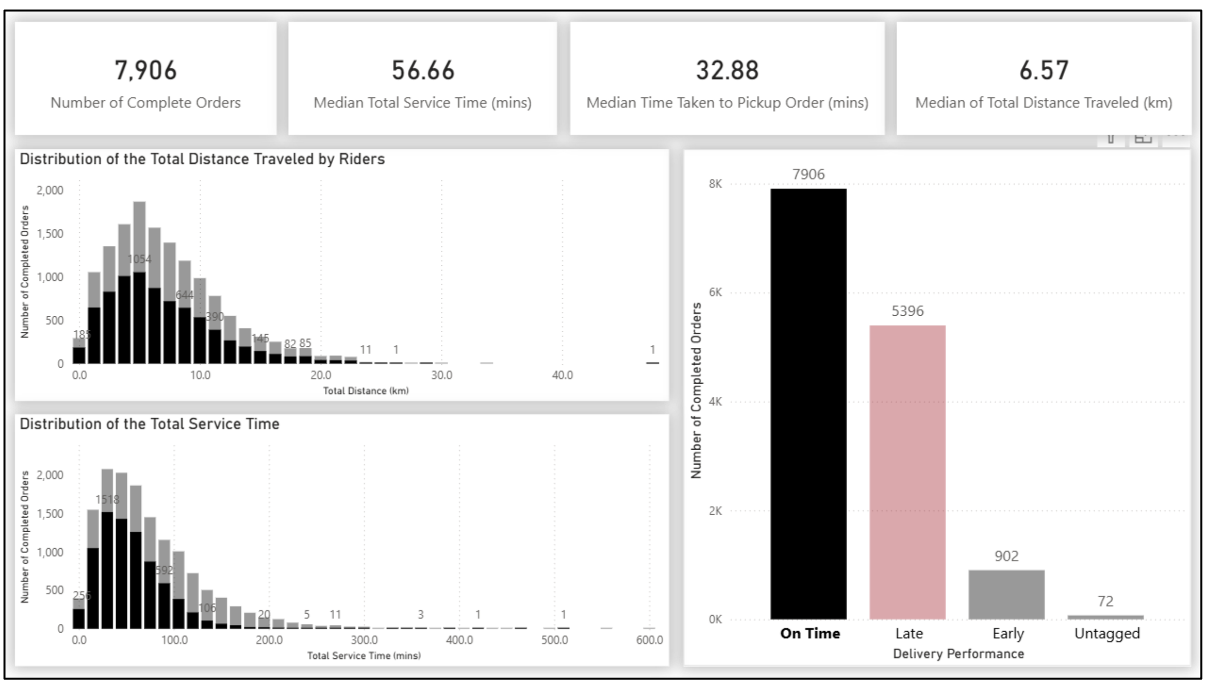


Completed Orders

If we look at the 5,396 late orders, the median total service time is as high as 106.58 minutes. A customer has to wait that long to receive an order. In the distribution of total distance traveled by riders, we see many customers living more than 15 km from the store, yet the median is only 7.78 km. It will not make sense to say that the longer the distance, the greater the tendency to be late in completing an order. As you can see, the median time for a rider to pick up an order is 71.34 minutes, accommodating more than 50% of the median total service time. The processes from the rider receiving a notification to pick up an order to picking up the order itself must have some inefficiencies.



For orders rated as on time, the median delivery time is 56.66 minutes, with a median total distance traveled by the rider of 6.57 km. This is a bit close to the median distance of being late (7.78 km)



Cancelled Orders

For the cancelled orders, 3 stages of the service are defined:

Stage 0: The customer places an order.

Stage 1: Orders are prepared and ready for pickup.

Stage 2: Orders are picked up and are in the process of being delivered.

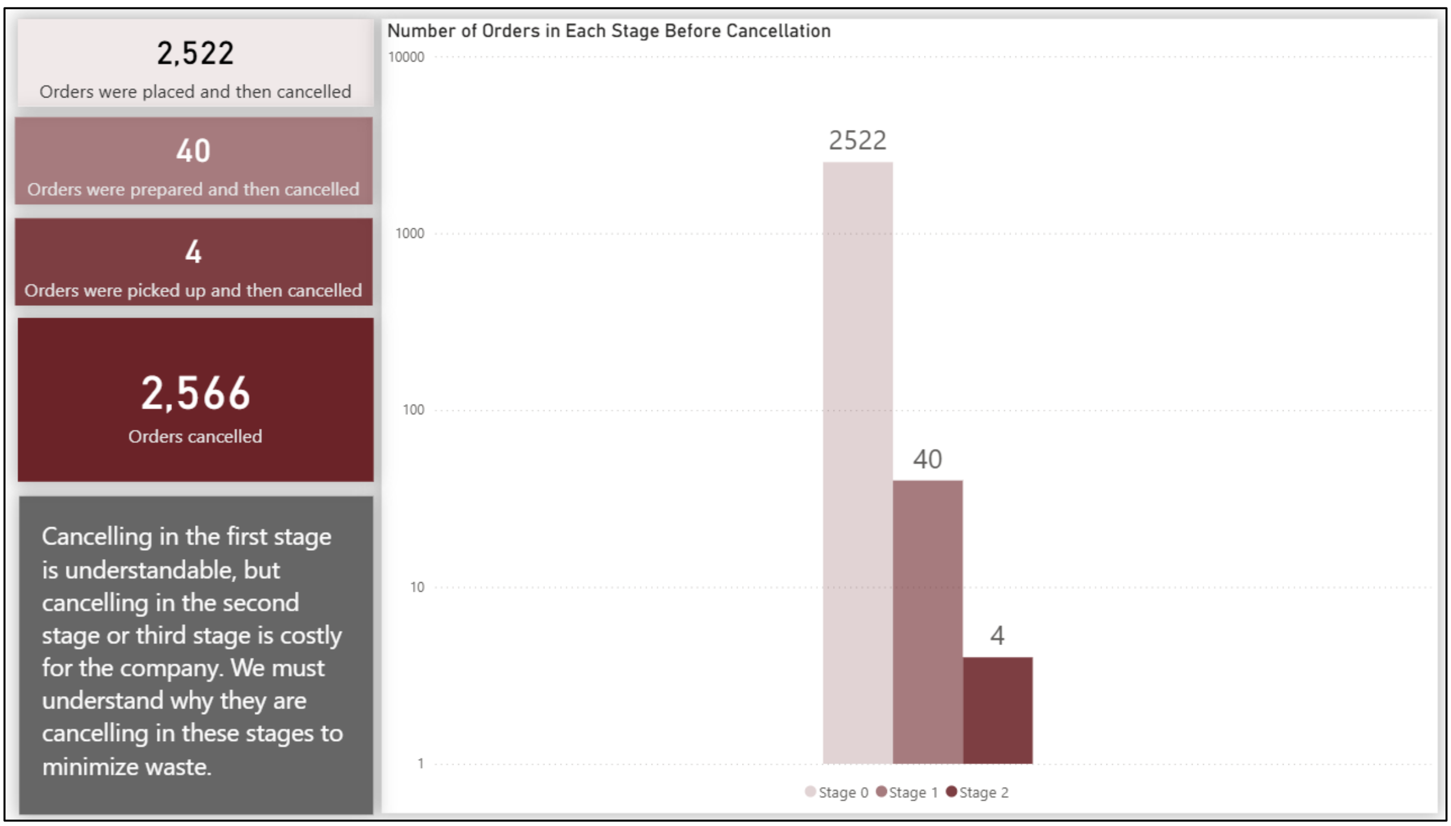
Stage 3: Orders are delivered, but then they will no longer be considered as cancelled.

When an order is cancelled, costs will be incurred. Resources are already being used up to prepare the order. The higher the stage achieved before being cancelled, the more expensive it will be for the company. Imagine the food has been prepared, picked up by the rider who has used up a lot of gas, attempts to deliver it to the customer while using up more gas, and then the order is cancelled halfway through.

2522 orders were placed and then cancelled. Food ingredients may already be used up, overhead may have increased, and the rider may already have been notified of the order and be on their way to the store.

40 orders were ready and then cancelled. All ingredients and utilities used to fulfill the order were wasted, and the rider may also have used some gas to pick up the order.

4 orders were picked up and then cancelled. All the effort it took for everyone involved in the order was wasted. Physical and financial resources were used up. All those resources are sunk costs.



Correlational Statistics

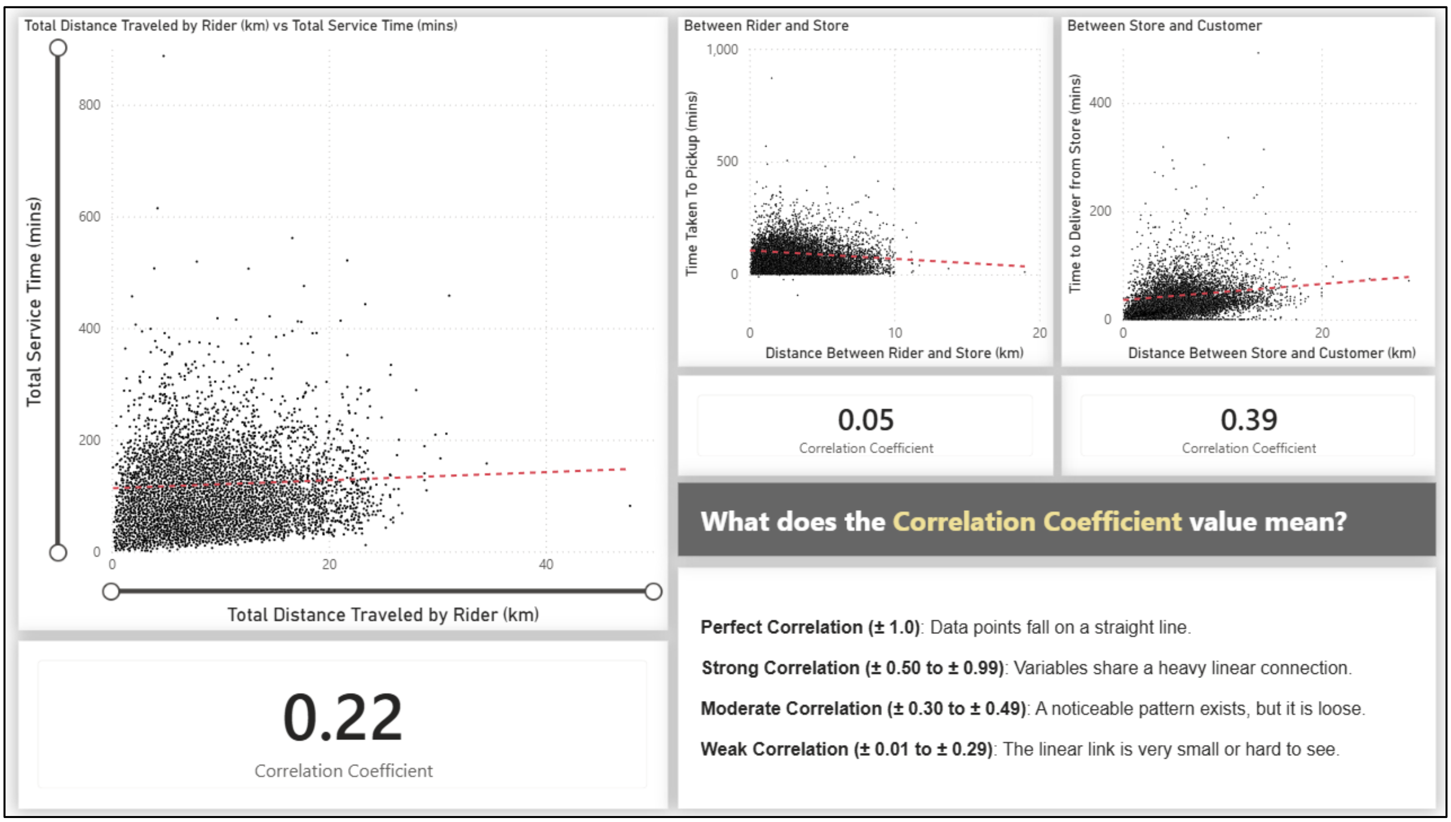
Using Power BI's quick measure of correlation between 2 variables, it is observed that service time and delivery distance have a weak-to-moderate correlation. The distribution of each line graph appears to be exploding in almost all directions, with most orders concentrated at around 0 – 200 minutes and 0 – 20 km.

Total Service Time plotted against the Total Distance Traveled by the Rider shows a correlation of 0.22, indicating a weak overall relationship.

The Time Taken to Pickup and the Distance Between Rider and Store are shown to have a very low correlation of 0.05. This value further suggests operational inefficiencies within these procedures. Problems such as late notifications, rider distraction, or other factors could explain the long time it takes to pick up orders.

The Time to Deliver from Store and the Distance Between Store and Customer show a moderate correlation of 0.39. It could be because the rider's sole focus is on delivery, and the only variable to consider is the distance between the store and the customer.

These results only prove that the company should not focus solely on reducing distance but rather improve its overall delivery system as well.



Recommendations

The data analysis reveals significant operational inefficiencies, primarily related to long pickup times, a high order cancellation rate, and a weak link between distance and service time. To address these issues and improve overall performance, the following recommendations are proposed:

1. Optimize the Order-to-Pickup Process

The analysis shows that the median time for a rider to pick up an order is 71.34 minutes, which accounts for over 50% of the total service time for late orders. The very low correlation (0.05) between rider-to-store distance and pickup time indicates that this delay is not primarily driven by distance but by process inefficiencies.

Implement a Dynamic Rider Assignment System: Develop an algorithm that considers real-time traffic, rider availability, and order preparation status to assign the most efficient rider for each order. This can help prevent riders from arriving too early and waiting, or too late after the order is ready.

Establish "Ready for Pickup" Notifications: Ensure that riders are notified or dispatched only after the store confirms the order_ready_for_pickup_time. This eliminates unnecessary waiting time at the store and prevents riders from arriving before the order is prepared.

Audit Rider Arrival Data: The missing data for rider_arrived_at_pickup_time is a significant issue. Implement a mandatory check-in process for riders upon arrival at the store via a GPS-enabled application to improve data quality and pinpoint where delays occur.

2. Address the High Rate of Order Cancellations

A cancellation rate of over 15% (2,566 orders) represents a major source of wasted resources and revenue loss. The analysis shows that costs escalate with the stage at which an order is cancelled.

Implement Real-Time Order Tracking and Proactive Communication: Provide customers with accurate, real-time order status updates. For delayed orders, proactively send notifications explaining the reason for the delay and offering the option to cancel before the order is picked up (Stage 1). This can help manage customer expectations and reduce the likelihood of a customer cancelling out of frustration later in the process.

Incentivize Riders and Stores for Reducing Cancellations: Develop a performance metric for riders and stores that tracks the rate of cancellations at different stages. Bonus structures or recognition programs can be introduced for teams that successfully minimize cancellations, particularly in the later stages (Stage 2).

Root Cause Analysis for High-Value Cancellations: The 4 orders that were picked up and then cancelled (Stage 2) represent the most significant loss. A detailed investigation into these specific cases is critical. Why did these customers cancel after the order was in transit? Was it due to an extended delivery time, a damaged item, or a change in customer plans? Understanding this is vital for preventing future occurrences.

Consider Charging a Cancellation Fee: For cancellations after the order has been prepared (Stages 1 and 2), the company could implement a policy charging the customer a partial fee to offset sunk costs (ingredients, rider compensation).

3. Improve System Reliability and Data Integrity

Missing values may be caused by "riders forgetting" or "a defective system", undermining the ability to make data-driven decisions.

Audit the Order Management System: Conduct a thorough audit of the system responsible for recording order statuses and timestamps. Identify and fix any bugs or usability issues that cause data not to be recorded. The system should be robust and user-friendly to minimize human error.

Require and Validate Data Entry: Make recording all required timestamps mandatory for a rider to close out an order. Implement validation rules within the rider's app that prompt them to enter missing information before proceeding.

Re-evaluate Data Handling Procedures: Null values are converted to median values for float data. Future analysis should still flag and investigate orders with missing data, as they often indicate a process failure that, once corrected, can significantly improve operations.

4. Enhance Operational Transparency and Performance Tracking

The current analysis uses historical data, but the goal should be to create a real-time dashboard for continuous improvement. Since distance is no longer a major concern, it will be dropped from the final dashboard.

Create a live dashboard using Power BI that tracks key metrics such as:

- **Order Fulfillment Rate (%)** vs. **Cancellation Rate (%)**.
- **Median Service Times (Pickup and Delivery)**
- **Order Status Funnel**: Track the order volume at each stage (Placed - Prepared - Picked Up - Delivered) to visually identify the stage with the highest drop-off/cancellation rate.

Proposed one-page dashboard: